

Welding

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Many people who begin to build a gasifier soon realize that they aren't very good at welding. A lot of this material is thin and may require extra attention even from experienced welders. Successful gasification depends on the absolute separation of fuel gases from fresh air. Any mixing and you will have at best a weakened gas and at worst an explosion. All welds need to be airtight. In addition you will be dealing with condensate (water) storage which needs to be welded tightly, and you are building equipment that needs to withstand thousands of heating and cooling cycles, which can weaken an inferior weld. You don't want a catastrophic failure because of one sloppy weld. So what methods can be used to ensure strong airtight welds?

- 1. Use quality welding tools.** As a novice welder you are probably tempted to buy one of those cheap Chinese welders and stick a reel of flux core wire into it. I know because **I was tempted too (/premium/welder)** - I bought the welder, used Lincoln flux wire and managed to glue some metal together. I also burned holes in all my shirts, spent weeks plugging up leaky welds with more welds, and eventually I ended up with a leaky gasifier. Surprise! Had I started with a quality welder and good wire I would have saved money in the long run. You will too! This isn't your last gasifier, you know.... it's only your first of many. Once you start you can never stop, it's a lifetime addiction.
 - Get a quality American made welder. Several good brands to choose from, Lincoln, Miller, Hobart, Esab. We have no preference. Wayne currently uses a Lincoln SP-175 Plus.
 - Get quality wire for your welder. No sense in skimping here, this is the least of your expenses. Many welder manufacturers supply wire for their machines.
 - Invest the money in an auto-darkening helmet. They are inexpensive and you will instantly be a better welder.
 - Don't forget the gloves, slag hammer, and wire brush. These are less critical on quality, but you still get what you pay for.
 - Keep your machine set up and in tune. Make sure the drive rollers are correct, use the right tip for the size wire, etc.
 - Have respect for your torch and cables. You don't want them to fail in the middle of a project. Keep them off hot surfaces, they could melt. Never hit the torch to remove slag, use a wire brush.
- 2. Go with a gas-shielded welder.** Not flux core. This will make a big difference in your weld quality. I built one gasifier using flux core. It splatters, makes too much slag, and doesn't produce as smooth and clean of a weld. The wire is more expensive, although you don't have to buy gas. The one advantage is on a windy day it can't be blown out like a MIG welder can. So make yourself a sheltered spot from the wind and use a gas MIG. You will do much better in the long run.

- Some welders can run both flux and MIG. This will allow you to try out flux wire if you like. I think you'll agree that its no good for our purposes.
- Shielding gas is available in bottles from welding suppliers. You can buy a large bottle and lug it around or a small one and refill it all the time. Wayne has both, refilling the smaller out of the larger.
- There are different mixes of Argon and CO2 available. You can try different ones and see what you like. Most folks use 75-25 Argon/CO2 mix.
- For shielding gas, a flow meter is needed - not to be confused with a pressure regulator.
- In calm environments the gas flow may be reduced for economy, but in windy places you need a strong flow of gas to protect the weld. On very windy days outdoor welding may be impossible.
- An interesting development is the MultiPlaz plasma welder and cutter, which runs on water and alcohol, and makes no fumes. No gas to buy. This is not a proven technology so we can't recommend it yet. Keep an eye on this promising development.

3. Use the right size wire and the right settings. There are different settings for different situations. On thinner material you want a low heat and a low wire speed. Thicker metals need more heat and more wire. Different wire thicknesses are good for different jobs. Take your time and figure out how the settings affect the weld quality. Once you know how it works you will be as confident in adjusting it as you are adjusting the volume knob when the radio is too loud.

- When you switch wire sizes be sure to change the tips and feed rollers to match the new wire. Otherwise you could have slippage.
- Too much or too little roller tension will affect wire speed. Too much distorts the wire, too little slips.
- There are three different style drive rolls for each diameter wire. V, U and Knurled. V = solid wire, U = aluminum wire, Knurled = inner shield (flux) wire.
- 87hp ESAB wire is available in .030, .035, .045 diameters and is by far the best I have used. This wire wets out good and has very low spatter.
- For very thin metal you'll want .023 wire. 16ga works fine with either .023 or .030. Move up in wire size as the material gets thicker.
- Increasing the heat affects penetration. Refer to your welding manual for recommended settings on various thicknesses.
- Adjust the wire speed to match the heat setting and your rate of travel. Set it so that you are not slowing down waiting for more wire but it isn't piling up on you either. As you add heat, you must also add wire speed, and vice versa.

4. Clean all surfaces before you start. Molten metal will not stick to rust. It won't stick to paint, or oil or dirt. The torch may be hot enough to burn off some of the dirt - but in the process you will lose your weld quality! Always start with shiny steel surfaces, dry and clean.

- Rust, paint & grease contaminate the weld puddle. Smoke from burning paint or grease even outside the weld area contaminates the shielding gas, causing porosity.
- Use a wire wheel or brush to clean painted or rusty surfaces. Alternatively you can soak rusty pieces in a solution of molasses and water, or diluted vinegar.

- Degrease metal surfaces using a chemical degreaser or by heating the metal with a torch. Do not leave brake cleaner on the part, it is flammable and the fumes are poisonous!
- Also clean the grounding point - electricity will not flow through rust or paint. A poor ground will cause you endless problems. Be sure ground clamps are in good condition and connect securely to the welder.
- Galvanized surfaces should be avoided. Burning zinc produces toxic fumes and isn't good for your weld either. If galvanized metal cannot be avoided, clean thoroughly with a wire brush or muriatic acid.

5. **Get the pieces to fit.** There's nothing as hard to weld as two pieces that aren't next to each other. While some types of wire can bridge the gap, the weld will be inferior quality. Cut your pieces square and straight, and clamp them tightly to get proper fit up. This will make your welding job much easier and your quality of work will go up significantly.

- Better fit up = smaller beads = less heat = less distortion.
- Joint preparation on thicker materials: Take a grinder and vee out butt welds. Use cabinet joint corners where ever possible.
- The better your fit things together the stronger the weld joint will be. A good fit will make it much easier to weld, especially for beginners.
- Wayne has been known to weld a bit, hammer the parts closer together, weld a bit more, hammer more, etc. This is a good way to form fit a flexible piece to an irregular hole.
- Sometimes it's enough to simply weight the pieces down or clamp them into a form. Be aware that any stresses built up while weighted down may be suddenly released when the form is removed. Be careful that you don't lock your form into place by welding!

6. **Go slow and smooth.** This is where the skill comes in. Have you ever tried drawing a straight line freehand? Slowly? Most people are a bit shaky with freehand work. Yet this control is essential to good welding. Fortunately you have many aids to steady you, and you can many times follow a groove or corner with your weld - but the speed of travel is very important. Too slow and you will burn through. Too fast and you will outrun your weld puddle, or leave sections unwelded. The smoother your motions the prettier the weld will be. You want a pretty weld so that you know by looking at it that it is strong and airtight. An ugly weld may have bad areas that are hard to see, unwelded spots and pinholes, and is much weaker overall. This is why you bought a good welder, cleaned surfaces well, and set the controls properly.

- Pull the trigger and touch the tip to the surface. You will immediately see an arc between the tip and the surface.
- Move slowly and steadily along. Watch closely, you will see the melted pool of steel ahead of you.
- Maintain the puddle at a given size. Speeding up makes it smaller, slowing down it gets bigger.
- For the "roll of dimes" look, pause or move slightly back every quarter inch or so. Then continue forward. This creates a row of freeze lines which looks nicer than a continuous smooth weld.
- If you have to stop, overlap with the last weld before you start again.

7. **Welding Techniques.** Many times welding a straight line is not good enough. You have to take into account that metal expands and warps when heat is applied. This can pull your joints apart before you can weld them, and cause your welded pieces to be poorly aligned. Here's some techniques commonly used:

- Step welding: Start a weld and continue for several inches. Jump ahead several more inches and begin welding backwards towards your previous weld. When you finish jump ahead again and weld back to your previous welds. This method distributes the heat more evenly and prevents some amount of warping.
- Tack welding: Make small welds just enough to hold the pieces together, and go all the way around until it is fully tacked. Then proceed with step welding or straight welding. This method allows you to position the parts for trial fitting as well as holding the parts securely while welding. Note that tacks can break under the stress of welding, so don't rely entirely on them.
- Weld opposite sides: Just like it says, you weld a few inches on one side, then a little on the opposite side. Continue until you have connected all sides. This keeps the pieces from distorting before they can be held in place.
- Weave welding: on heavier pieces where you want a wider bead, you can weave the torch back and forth in a crisscross pattern, overlapping the previous welds. This makes a wider weld and a longer dwell time, leading to deeper penetration.

Welding takes a lot of practice. But you've got plenty to practice on! And it's a skill that will serve you well in many projects to come. Why not get started the right way!

About Drive On Wood

A community of gasification enthusiasts hosted by Wayne Keith, resident expert & woodgas pioneer.

Drive On Wood is a partnership of Wayne Keith & Chris Saenz.

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